Verification and Validation in Scientific Computing

Daniel Topa
daniel.m.topa.ctr@mail.mil

Defense Threat Reduction Information Analysis Center (DTRIAC)

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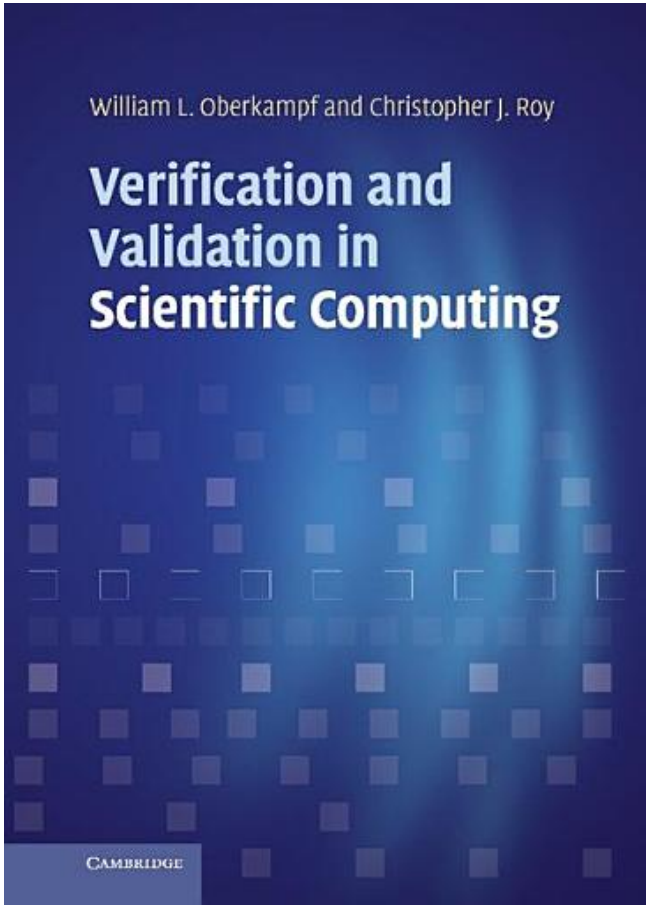
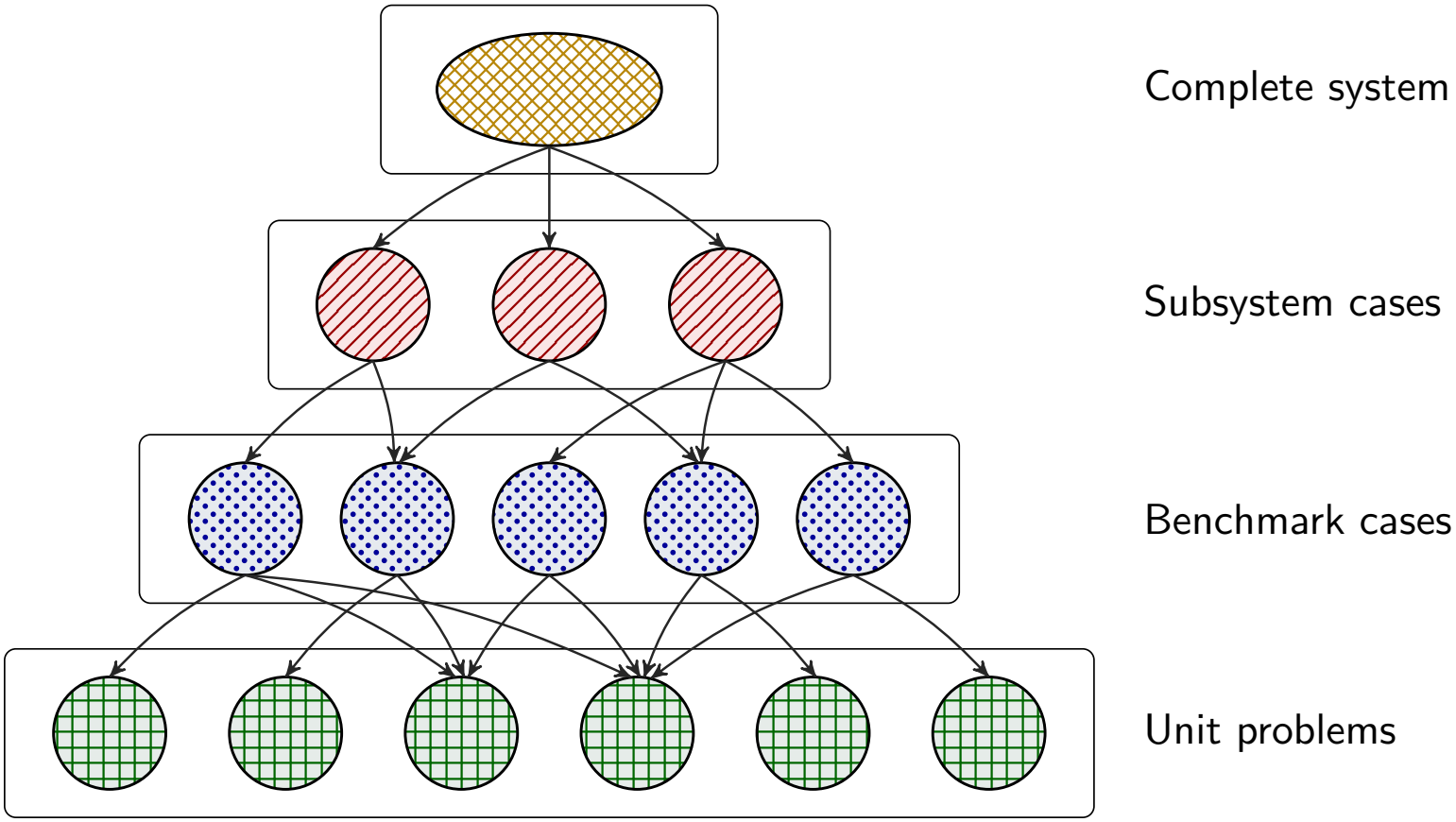
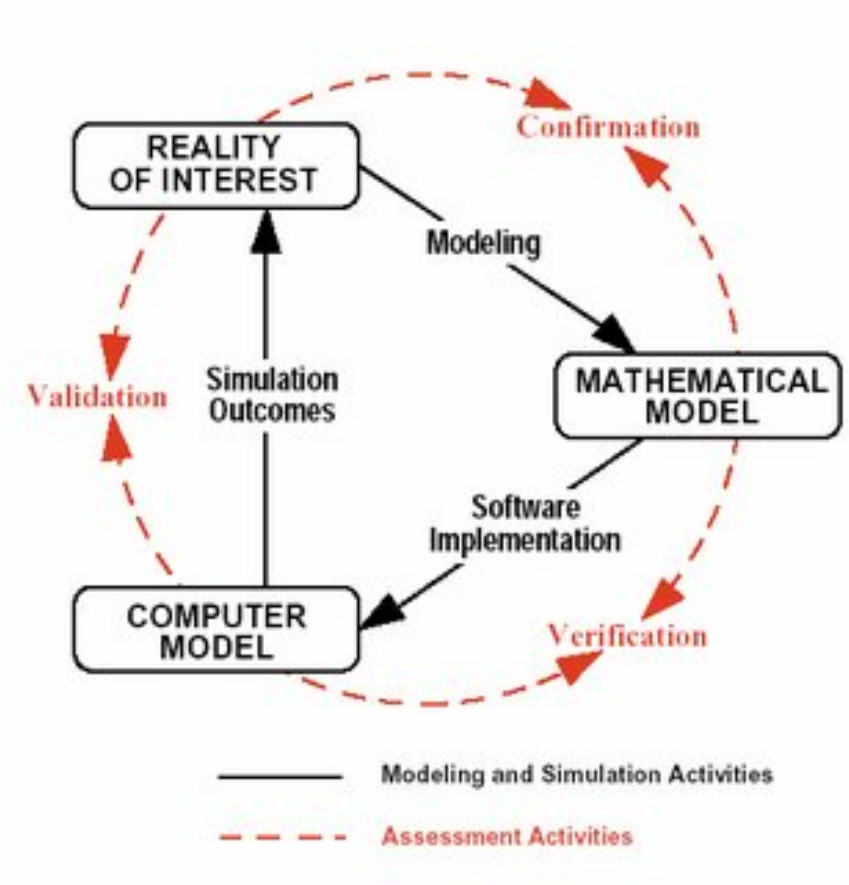


McAninch Validation Hierarchy



Phases of Modeling and Simulation



Overview

Fundamental Concepts

Code Verification

Solution Verification

Model Validation and Prediction

Planning, Management, Implementation

In the Very Beginning....

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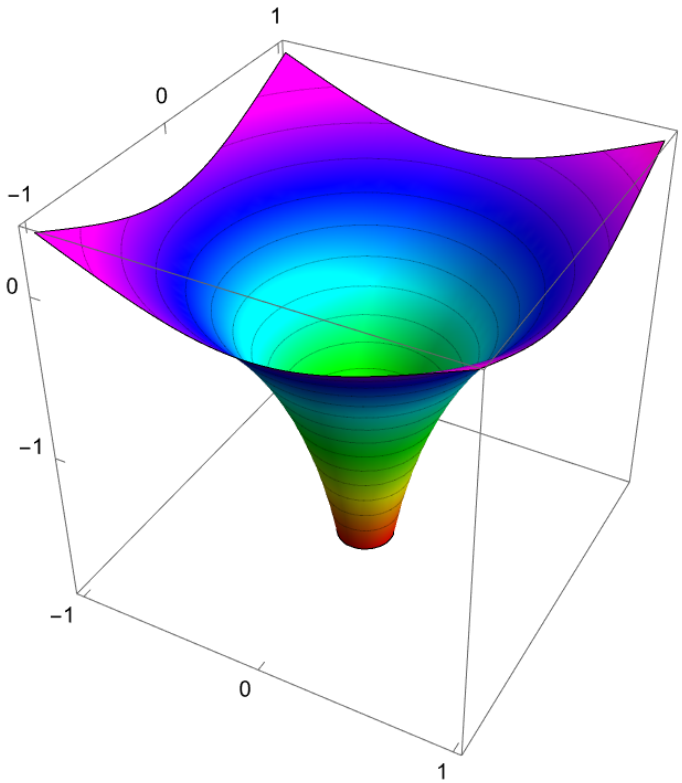

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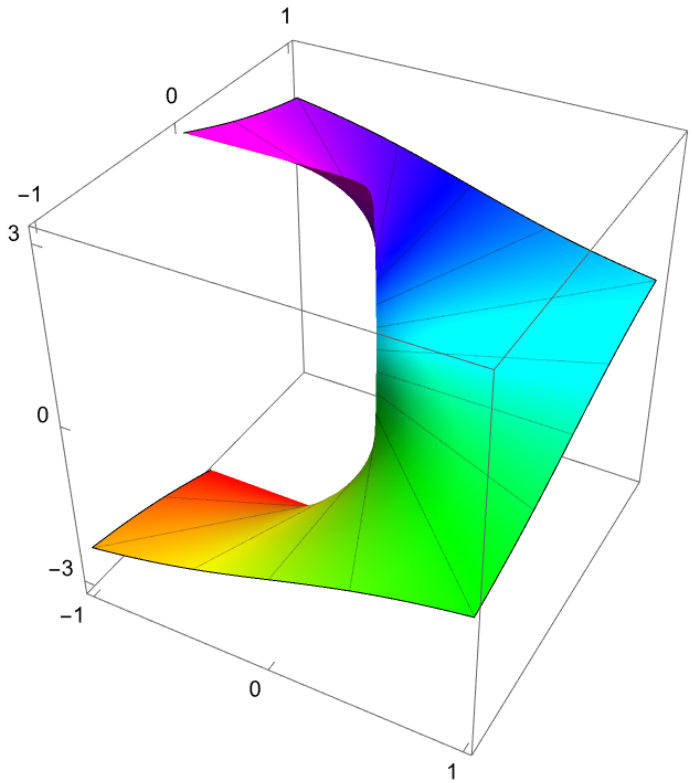
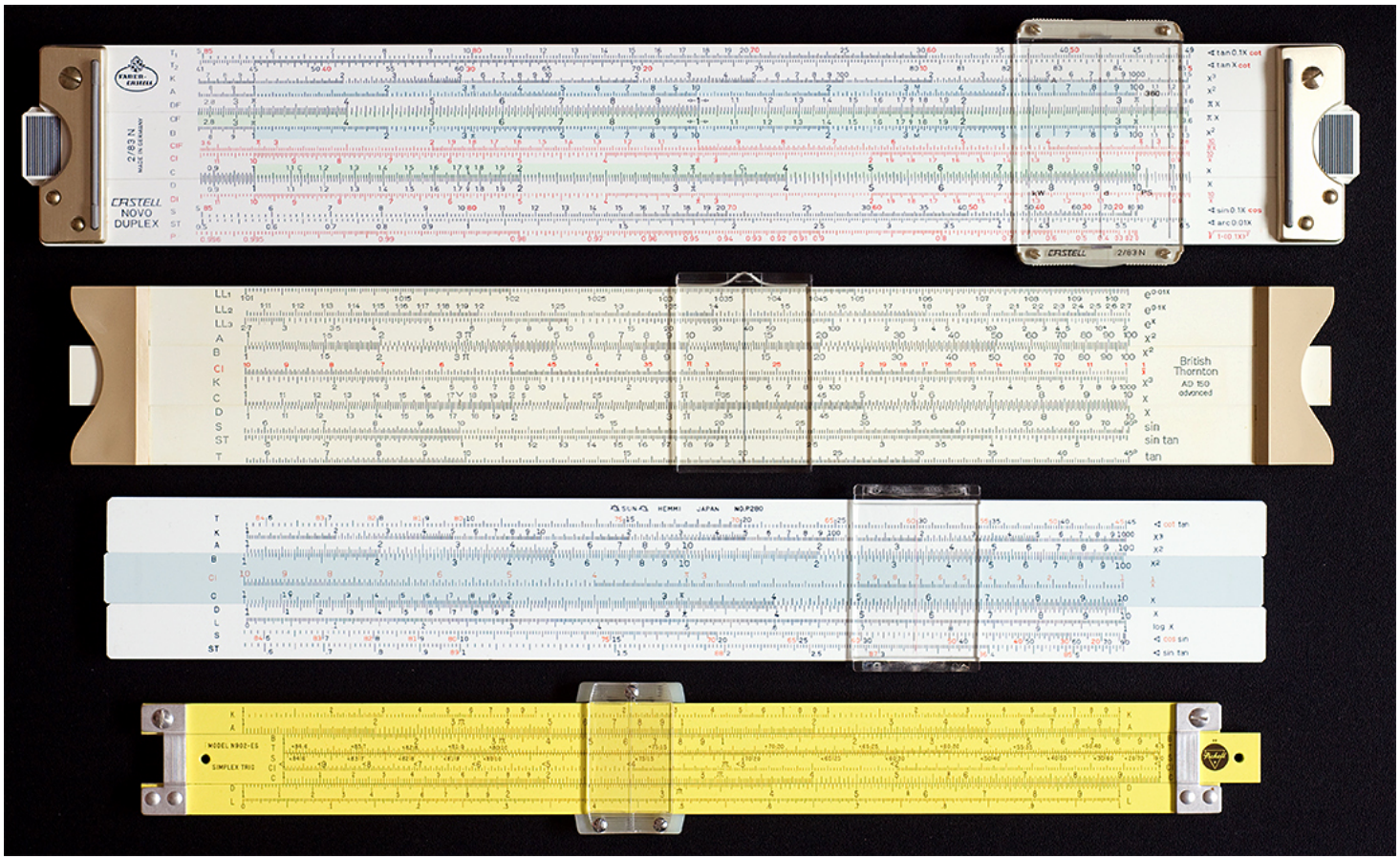


In the Very Beginning....

Property	Definition	Example
Product	$\log_b mn = \log_b m + \log_b n$	$\log_3 9x = \log_3 9 + \log_3 x$
Quotient	$\log_b \frac{m}{n} = \log_b m - \log_b n$	$\log_{\frac{1}{4}} \frac{4}{5} = \log_{\frac{1}{4}} 4 - \log_{\frac{1}{4}} 5$
Power	$\log_b m^p = p \cdot \log_b m$	$\log_2 8^x = x \cdot \log_2 8$
Equality	If $\log_b m = \log_b n$,, then $m = n$.	$\log_8(3x-4) = \log_8(5x+2)$ so, 3x – 4 = 5x+2

In the Beginning....

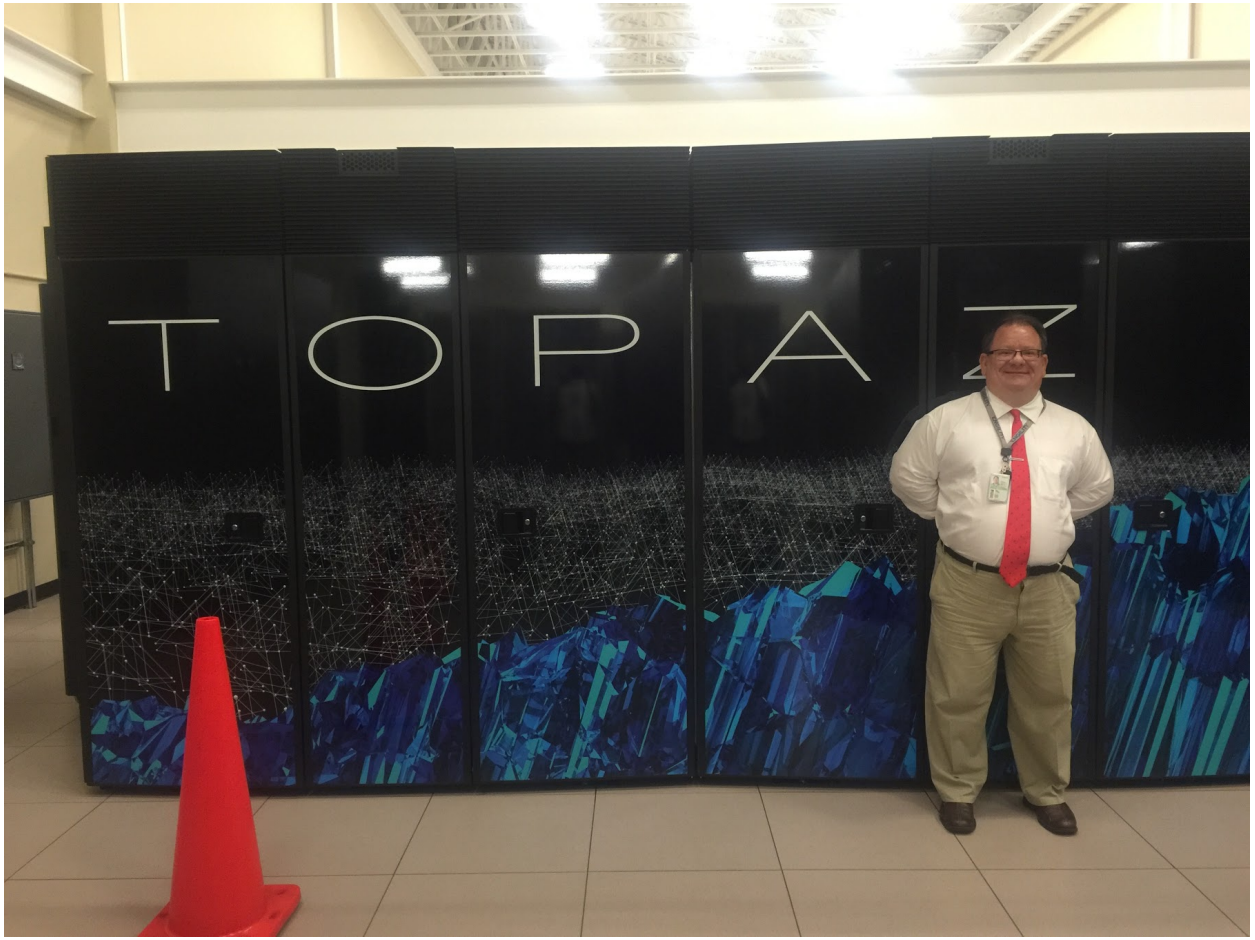


A Brief Timeline of the Slide Rule

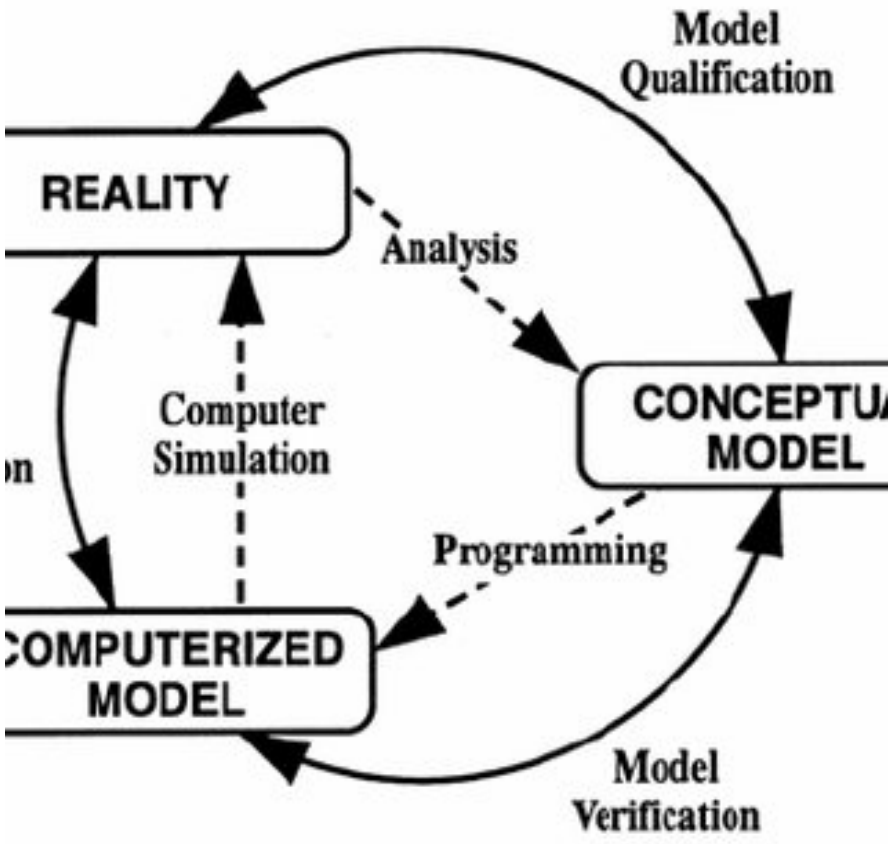
1614 - Invention of logarithms by John Napier, Baron of Merchiston, Scotland.
1617 - Developments of logarithms 'to base 10' by Henry Briggs, Professor of Mathematics, Oxford University.
1620 - Interpretation of logarithmic scale form by Edmund Gunter, Professor of Astronomy, London.
1630 - Invention of the slide rule by the Reverend William Oughtred, London.
1657 - Development of the moving slide/fixed stock principle by Seth Partridge, Surveyor and Mathematician, England.
1775 - Development of the slide rule cursor by John Robertson of the Royal Academy.
1815 - Invention of the log log scale principle by P.M. Roget of France.
1850 - Amédée Mannheim, France, produced the modern arrangement of scales.
1886 - Dennert & Pape, Germany, introduce white celluloid as a material for inscription of scales.
1890 - William Cox of the United States patented the duplex slide rule.
c1900 - Engine divided scales on celluloid increases precision of slide rules.
1976 - The final slide rule made by K&E donated to the Smithsonian Institute, Washington, DC, USA.

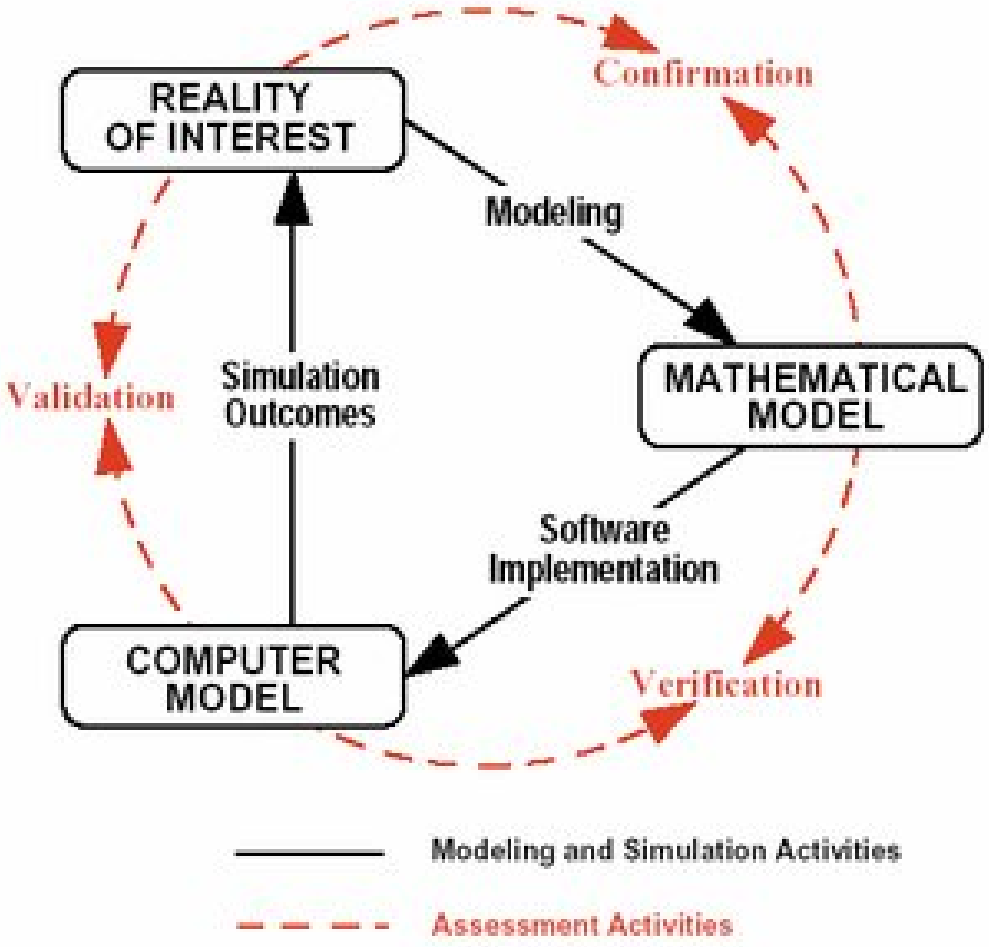
Programmable Calculators

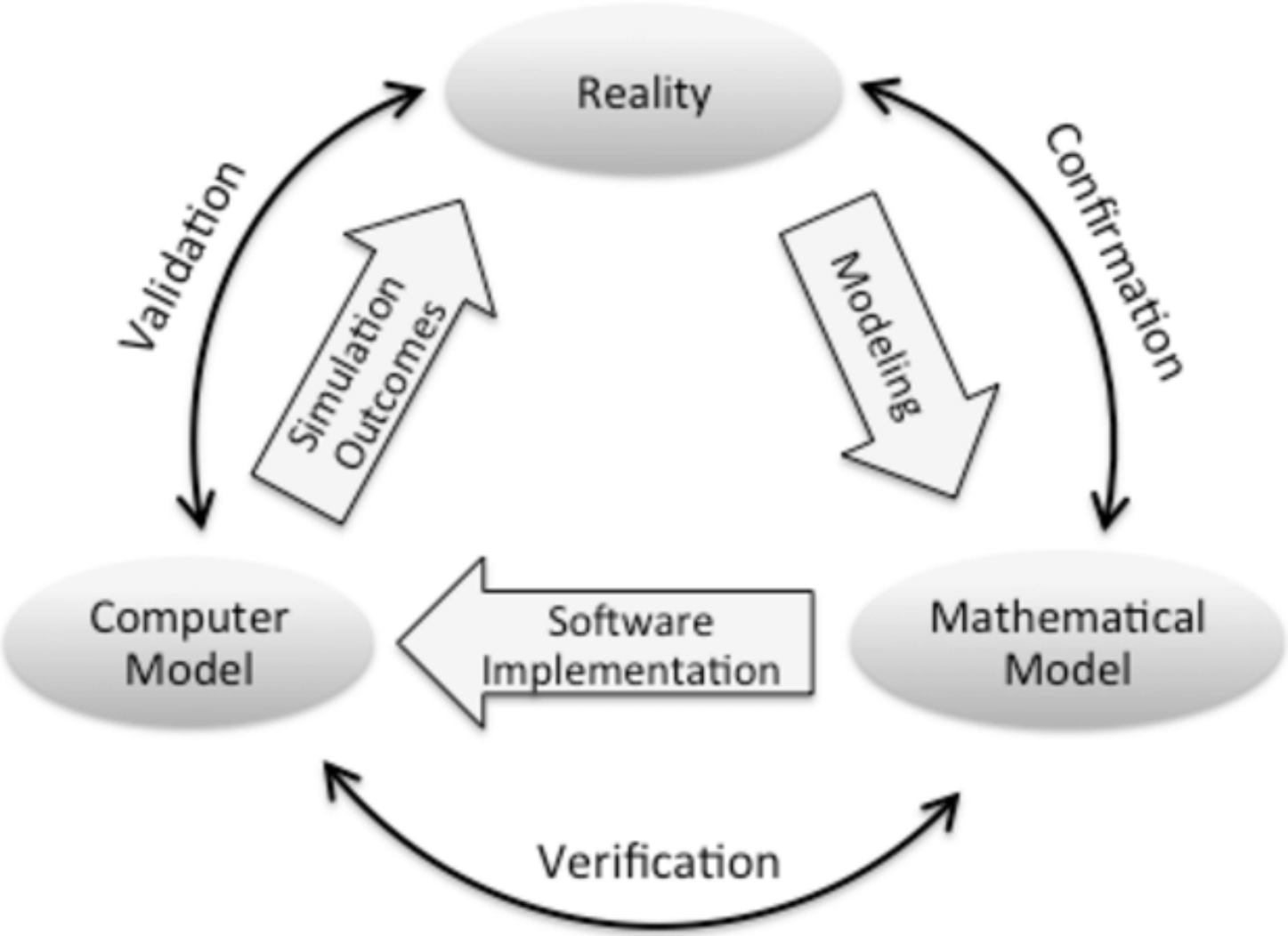
Supercomputers



Phases of Modeling and Simulation




REALITY OF INTEREST
Confirmation
Modeling
MATHEMATICAL MODEL
Validation
Simulation Outcomes
Software Implementation
COMPUTER MODEL
Verification
Modeling and Simulation Activities
Assessment Activities


Reality
Validation
Confirmation
Simulation Outcomes
Modeling
Computer Model
Software Implementation
Mathematical Model
Verification

Personae Dramatis

- DoD: Department of Defense
- ASME: American Society of Mechanical Engineers
- AIAA: American Institute of Aeronautics and Astronautics

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